Techcrast
CONNECTEDSUITE

BUILD PLAN · EVERY SYSTEM, FOR AXE AND MICROMART

Powering Up the Front Doors

What it takes for every system in the estate to open, in each partner's own brand, for Axe Capital and Micromart: what is live, what was built on 20 October, what is dark, and the order to light the rest. The BirgenAI wallet moves to connectedsuite.servicesuitecloud.com, beside the LMS on servicesuite-in.

For	Date	Repos	Host
Founder and engineering	**20 October 2026**	**connected-suite · birgen-ai-frontend · micro-eazy-app · interchange**	**servicesuite-in (213.148.17.228)**

Internal. Code described as built is in the working trees, uncommitted and not yet deployed, unless marked live.

01 Where every door stands today

A "front door" is everything between a person and a system: the host name that resolves, the branded sign-in card in the lender's livery, the installable app, the landing in the right system, and the system reading that lender's real data. Checked on 20 October 2026:

SYSTEM	HOST	AXE CAPITAL	MICROMART
Lending Console	lms.	Live Boresha and Stawi on Axe's ServiceSuite (live 19 Oct)	Live entities 3002 and 3005
CallBox	callbox. / connectdesk.	Live ConnectBox floor, Linkus calling	Partial floor live; Request payment not wired
Customer Portal	microeazy.	Not built Axe locked in the lender chooser	Dark host has no DNS record
Analytics	analytics.	Verify against Axe's schema gaps	Live
Ledgerly	ledgerly.	Verify Axe's journal	Live
PeopleHub	peoplehub.	Verify Axe's staff directory	Live
The Interchange	interchange.	Dark host has no DNS record; members AXE-3003/3004 active	Dark same
BirgenAI wallet (Hub)	birgenai.com	Dark birgenai.com has no DNS (NXDOMAIN). Every "Pay with M-PESA" answers "wallet not connected".	

lms., callbox., connectdesk., connectedsuite., analytics., peoplehub. and ledgerly. all resolve to 213.148.17.228 and answer 200. microeazy., interchange. and portal. return no address (checked through 1.1.1.1). Both were Vercel projects, and the Vercel team has been paused since 14 October.

02 Built on 20 October (uncommitted)

Billing: the estate price book and downloadable invoices — connected-suite

FILE	WHAT IT DOES
`src/lib/billing/catalogue.ts`	Per-system list prices, the multi-system discount (10/15/20%), the Startup edition, and partner agreements in code: **Axe TC-FPA-2026-02** (console + CallBox + Interchange = 30,000) and **Micromart TC-FPA-2026-01** (whole estate = 30,000). An organisation with no agreement and `systems = null` is billed exactly as before.
`src/lib/billing/invoice.ts`	The month-end freeze and the open-month estimate now price through the catalogue: one line per system, then the discount or partner line, then usage.
`src/lib/billing/issuer.ts`	Techcrast as issuer; Till 5620596 as payee. Both can be changed by environment variable.
`src/lib/billing/invoice-pdf.ts` + `api/console/billing/invoices/[id]`	The invoice as a Techcrast-letterhead PDF (pdfkit), View and Download.

`api/console/billing` + `console/billing/page.tsx`	Systems breakdown; **Pay** per invoice through the wallet; **I paid the till** records an M-PESA code; partners are not offered a package switch; only Startup (20k) and Premium (30k) are on sale.
`src/lib/billing/hub.ts`	Signs an **invoice token** (number + amount inside the HMAC) and reads paid invoices back from the Hub. Fixes a latent bug: the sync wrote the Hub's paid-through date into the billing month (`currentPeriodEnd`), which would have shifted every invoice to the payment date. Default Hub URL is now connectedsuite.servicesuitecloud.com.
`api/platform/orgs` + `PlatformBoard.tsx`	**Mark paid** (with the till's M-PESA code) and **Void** on each unpaid invoice.
`sql/billing-2026-10-zoho-handover.sql`	Axe to PREMIUM; voids the five in-app invoices that duplicate the Zoho invoices (Axe Aug–Sep, Micromart Jul–Sep); imports October as INV000088 (Axe, due) and INV000089 (Micromart, paid). Rehearsed and rolled back.

The wallet: birgen-ai-frontend

- `lib/billing/lms.ts`: `verifyLmsInvoiceToken` (signature, expiry, invoice number, amount capped at `LMS_INVOICE_MAX_KES` = 500,000).
- `api/payments/stk-push`: charges the signed invoice amount, settled through the existing LMS subscription path (+30 days).
- `api/lms/subscription`: also returns the invoices settled through the wallet, so the LMS marks them PAID.
- `transact/page.tsx`: an invoice card, and the return link now accepts `*.servicesuitecloud.com`. Until today it bounced every lender back to birgenai.com.
- `ratecard.ts`: Advanced renamed Startup (same key and price). `.env.example`: till 5620596, Buy Goods, callback on connectedsuite.

Each lender installs its own app — connected-suite

- `/<slug>/manifest.webmanifest`: the lender's own manifest (name "Axe Capital", Axe's mark, Axe's theme colour, its own `id`, start URL its door, shortcuts only to systems it holds). Verified locally for axe, fastdial and an unknown slug (404).
- The lender's door, the shared `/login` (when the browser remembers the lender) and every signed-in staff page (`staffTabMeta`) link that manifest.
- `InstallAppButton` on both door layouts: "Install Axe Capital" opens the browser's install sheet; iPhone gets Add-to-Home-Screen guidance; hidden once installed.
- `scripts/make-lender-app-icons.ts` made 192, 512 and maskable icons for axe, micromart, mular and fastdial.

03 The wallet at connectedsuite.servicesuitecloud.com

> **That host is in use today**
> connectedsuite. is bound to the LMS site (LmsNode, IIS site 3) and on its certificate. It was the stopgap staff used on 19 October when Google's DNS cache broke callbox. Before the Hub takes it, move anyone still using connectedsuite.servicesuitecloud.com/<slug> to lms. or callbox., and remove it from the host list in `bridge/lms-node/common.ps1`.

#	STEP	WHO
H1	Database: create `birgenai_hub` on LmsPostgres :5433 with roles `hub_owner` / `hub_app`. Apply the Hub's Prisma schema offline (`prisma migrate diff --from-empty`, then apply through `pg`; `db push` fails here). If the old Supabase Hub data matters (users, wallet balances), copy it first.	Founder (pgAdmin)
H2	Build: a standalone Next build of birgen-ai-frontend in GitHub Actions, deployed by the same runner, as `C:\HubNode` on port 3200 under a service `HubWeb`. Copy the lms-node kit: `HOSTNAME=localhost`, slim pack, symlinks materialised.	Engineering
H3	IIS: remove connectedsuite. from LmsNode; a new site HubNode bound to connectedsuite.servicesuitecloud.com on 443; ARR to localhost:3200; its own win-acme certificate.	Founder on .228
H4	Daraja for Till 5620596: a Daraja production app with Lipa Na M-PESA Online for the till's **store (head-office) number**. Safaricom issues the passkey for that store. Set `MPESA_ENVIRONMENT=production`, `MPESA_BUSINESS_SHORTCODE=<store>`, `MPESA_TILL_NUMBER=5620596`, `MPESA_TRANSACTION_TYPE=CustomerBuyGoodsOnline`, consumer key, secret and passkey, and the callback at https://connectedsuite.servicesuitecloud.com/api/payments/callback.	Founder with Safaricom
H5	Shared secret: one random `LMS_BILLING_SECRET`, set in the Hub env and appended to `C:\LmsNode\lms.env` together with `HUB_BILLING_URL=https://connectedsuite.servicesuitecloud.com`, then `Restart-Service LmsWeb`. Use `[IO.File]::AppendAllText`, never PowerShell 5.1 `Set-Content`.	Founder on .228
H6	Proof: a KES 10 test invoice on the demo organisation; pay it from the console with a real STK push; confirm the receipt on the till statement, then the invoice PAID after the nightly sync (or the sync button).	Both

Known friction, next to fix. The Hub requires its own account before it will open an STK push (`PaymentIntent.userId`). A lender's billing admin must therefore sign up to the wallet once. The fix is to let a valid LMS invoice token stand in for that: the Hub finds or creates a payer account from the staff email the LMS signs into the token. That makes BirgenAI ID reach the wallet.

04 Billing go-live, in order

1. **Before 1 Nov 03:00 EAT:** run `sql/billing-2026-10-zoho-handover.sql` in pgAdmin on production. If it runs later, the cron freezes its own October invoice first, and Axe and Micromart are billed twice for October.
2. Commit and deploy connected-suite (the workflow deploys to .228 when `LMS_NODE_DEPLOY=on`). Commit only the files listed in section 2.

3. Check, as each partner's admin, that Billing shows 30,000 and lists INV000088 / INV000089 with working PDFs.
4. Until the wallet is live (section 3), lenders pay Till 5620596 and press **I paid the till**; you confirm on the platform board with **Mark paid**.
5. 1 December: the first ConnectedSuite invoices, INV-AXE-202611 and INV-MICROMART-202611, at 30,000 each.

Two gaps to close in code. (a) The price book promises 14 days' grace before metered services pause, but the Hub sync marks a lender past due as soon as the wallet says unpaid; add the grace rule in `syncSubscriptionFromHub` before the wallet goes live. (b) eTIMS: lenders need KRA eTIMS invoices to deduct the expense; connect the invoice freeze to eTIMS, or issue the eTIMS invoice by hand alongside it.

Side effect to confirm. Fastdial holds only CallBox (`systems = ["callcenter"]`), so it now prices at 15,000 instead of Premium's 30,000. That matches the price book. Confirm it is intended before 1 December.

05 Front-door workstreams

FD1 · Light the dark hosts (first, this week)

- **microeazy.** is Micromart's customer app (micro-eazy-app, a static Vite SPA whose `/api` proxies to lms.). Build it to static files, serve it from IIS on .228 as its own site, rewrite `/api/*` to LmsNode, add an A record and a certificate. Static, so it is the easiest of the three.
- **interchange.** is the Interchange console (Next). Deploy it beside the engine on .228 the way the LMS is deployed, with `INTERCHANGE_*` variables copied from the Vercel set of 22 September. Use `dotenv.parse`: `INTERCHANGE_NODE_KEYS` is single-quoted JSON.
- **portal.** was Micromart's own PWA. Confirm with Micromart whether it is still wanted before restoring it.

FD2 · Axe's Customer Portal

- micro-eazy-app: mark Axe `available` in `src/lib/lenders.ts`. Its livery comes from the LMS `Org` row (navy and orange since the 13 Oct rebrand).
- The app shows only LMS products, so finish Axe's shelf import in the console first.
- Repayment: Axe's Daraja keys in Axe's console vault (Settings & Vault), so "Pay now" pushes STK to Axe's paybill, never to the BirgenAI till.
- Axe's customers' installable app: the same per-lender manifest approach as staff (section 2), in micro-eazy-app.

FD3 · Analytics, Ledgerly and PeopleHub on Axe's books

Axe's ServiceSuite lacks `DateCleared`, `ProvisionRates` and `sp_NewBorrowerRegistration` (schema-caps layer, 19 Oct). For each of the three systems: open it as an Axe admin; list every panel that errors or reads empty; route each through schema-caps or hide it with an honest "not in your book" note; then demonstrate it to Axe's product lead under the co-development clause.

FD4 · CallBox for Micromart

Wire **Request payment** (STK from the case file) to Micromart's Daraja app on entity 3005, through Micromart's RepaymentTrigger allocation, and confirm the CollectBox mirror before arming any write.

FD5 · Every door branded, every app installable

- Door artwork exists for Axe (lms and callcenter) and Micromart. Add analytics, ledgerly and peoplehub door art per lender in `lib/suite/doors.ts`, or let those doors fall back to the plain branded card.
- Replace Axe's counter photograph, whose wall banner shows Micromart's slogan (flagged 15 Oct).
- Test the install on Windows Chrome and Edge, Android Chrome and iPhone Safari from `lms.servicesuitecloud.com/axe` and `/micromart`. Check the desktop shortcut reads "Axe Capital" with Axe's mark and opens Axe's door.

FD6 · BirgenAI ID across the estate

Staff sign in once across the six LMS-hosted systems today (shared session cookie). The Interchange authenticates members by node key, and the wallet by its own account. Next: the wallet trusts the LMS invoice token as identity (section 3), and the Interchange console accepts a BirgenAI ID session mapped to the member's organisation, so an Axe credit officer opens the Exposure Check without a second sign-in.

06 Sequence

WHEN	MILESTONE	DONE MEANS
By 31 Oct	Billing handover	SQL run; billing code deployed; both partners see 30,000 and download their October invoice
Week of 27 Oct	FD1 dark hosts	microeazy. and interchange. answer from .228 with valid certificates
Early Nov	Wallet live	H1–H6 done; a real STK payment to Till 5620596 marks an invoice paid
Mid Nov	FD3 + FD5	Axe opens Analytics, Ledgerly and PeopleHub on its own books; apps installed on both partners' staff machines
Late Nov	FD2 + FD4	Axe customers apply and repay in the Axe app; Micromart agents request payment from the case file
1 Dec	First estate invoices	INV-AXE-202611 and INV-MICROMART-202611 issued, paid through the wallet
December	FD6	One sign-in reaches the wallet and the Interchange

07 Your steps, collected

#	ACTION
1	Run the billing handover SQL in pgAdmin before 1 Nov 03:00 EAT.
2	Decide whose name the invoices carry, given that the money lands in Till 5620596 (see the Founder's Control Guide, section 2).
3	Daraja production credentials for Till 5620596's store number (H4).

4	Create the Hub database (H1); IIS and certificate on .228 (H3); env and restart (H5).
5	DNS A records for microeazy. and interchange. at cloudoon.
6	Confirm Fastdial's 15,000 CallBox-only price, and Micromart's 30,000 whole-estate rate.
7	Metropol's written approval of Interchange bureau fulfilment for Axe.
8	Move the invoice PDFs and the old guide out of `connected-suite/public/lenders/`; anything committed there is served publicly once deployed.